

Readership and Gender in Today's America

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Abstract—This study aims to explore the trends in readership and book genres in the U.S. and investigate their relationship with gender as it pertains to today. By using various data on book sales, genre trends and demographic reading data on gender, this research explores the aspects of readership over time and looks at how gender is connected to them. Through data visualization and analysis, this study explores the reasoning behind the gender gap in readership and the difference in genres read according to gender. Notably, this study reveals the driving sociocultural and technological forces behind this gap.

Keywords—Print Book, E-book, TikTok, BookTok, Genre

I. INTRODUCTION

This project looks at the way the relationship between gender and readership has changed over time in the United States. This study focuses on specific factors such as books sold, technological impacts, genre trends and gender distributions across genres in addition to major sociocultural changes through the past 150 years to explain the current gendered readership trends.

Literature has always been closely tied with societal changes, reflecting social problems such as war, poverty and injustice. Religion, historical events, philosophy and psychology are only some of the many examples of literary topics throughout history and all reflect the way people view the world and each other. In addition to these topics of nonfiction, fiction has also been an important genre. Fiction and fantasy genres show parallels to our world and society in a way that is engaging and accessible. Altogether, writing and literature are important aspects of civilization and human culture.

This study looks at the gender distribution in regard to readership (books read) and genre. By using various demographic data alongside historical and sociocultural events, this research project reveals the relationship between gender and readership in America today. This data includes the number of books sold over recent decades, the impact of technology on reading, the trend of genres over time and the gender distribution of genders today. This demographic data shows that there is a difference in the types of books men and women consume, as well as how many books.

II. DATASETS

A. Source of datasets

The first dataset, “Unit sales of print books in the United States from 2004 to 2020”, sourced from Statista, tracks the number of print books sold from 2004-2020 [4]. The second, “Estimated number of e-books sold in the United States from 2010 to 2020”, also sourced from Statista, shows the number of E-Books sold from 2010-2020 [3]. These were both compiled by a media researcher who looks at the fast-paced, ever-changing media landscape and how it impacts traditional formats. In this study, these datasets were used to look at the relationship between these formats and readers.

The third dataset, “Share of TikTok users reading more books because of #BookTok in the United States as of May

2023, by state” from Statista, looks at another technological impact on readership: the social media app TikTok [7]. This dataset looks at how many Americans in 42 states, are reading more because of a subcommunity on this app called BookTok.

The fourth dataset, “Best Books Ever”, sourced from Kaggle, is a comprehensive list of popular books published between 1916-2017 [13]. This study looks at how the trend of genres has evolved over time by using the most popular books from these years. Lastly, the fifth dataset, “What types of books have you read in the past year?”, from Statista, looks at the gender comparison of readers categorized by genre according to a survey done in July 2015.

B. Character of the datasets

The first and second datasets regarding books sold each have two fields; year and number of books sold. These datasets were also combined to be used for a model by merging each of their number of books sold columns and using a range of years included in both sets. Table 1 shows the fields of the data for this third model.

TABLE 1. Total Books Sold data format

Field Name	Description	Format
Year	A year books were sold	YYYY
Print books sold	The number of print books sold per year	Integer
E-books Sold	The number of e-books sold per year	Integer
Total books sold	The sum of print and e-books sold per year	Integer

The third dataset contains a column of states and a column of the percentage each state is reading more because of BookTok. The fourth dataset had numerous, comprehensive fields for over 50,000 books. However, not all were necessary for this study. To look at the trend of genres, the variables used were “bookId”, “genres” and “firstPublishDate”. The year was extracted from firstPublishDate to then count each genre that appears in “genres” per year. Table 2 displays the fields used for the purpose of this study.

TABLE 2. Best Books Ever relevant data format

Field Name	Description	Format
bookId	The numeric ID of the book followed by its name	XXX.name
genres	The genre(s) the book is a part of as a list	['genre1', 'genre2'..., genreX]
firstPublishDate	The date the book was first published	Date (MM/DD/YYYY)

To track the trend of genres per year, each genre a book was a part of had to be counted. Data cleaning was performed to go through each book's list of genres and strip out each genre, putting it in its own list. Then, a count was established for each genre. Additionally, any rows of data that were empty or duplicated (same book and year) were removed.

The fifth dataset, regarding book genres by gender, has three fields; an unnamed column referring to genre, a column for male readers, and a column for female readers, both in percentages. The male and female reader percentages refer to the percentage of each gender reading books in the specified genre according to the responses in the survey.

III. METHODOLOGY

A. Data Visualization and Analysis Methods

To conduct data visualization, data cleaning and analysis were performed using Python in Visual Studio Code using Jupyter Notebook. Libraries include numpy, matplotlib, pandas, seaborn and ast. The aim of these tools was to provide clear visualizations that create insights into the topic of this study.

Tools and Techniques used:

- Visual Studio Code: VS Code was used to establish a Python environment for the Jupyter Notebook used for data cleaning and visualization.
- numpy: Used for numerical operations, specifically on arrays and matrices.
- pandas: Allows for ease of data cleaning and manipulation.
- matplotlib: Used to plot data in various formats in order to visually assess trends and distributions.
- seaborn: Another library used for graphs that has a higher-level interface and design aspects.
- ast: A library that works with Abstract Syntax Tree of Python source code. For this project, it was used to evaluate strings contained in a list for one of the models.

B. Models

Various models were used to appropriately visualize the different datasets. Regarding tracking the trend of books sold, scatter plots were used as this data is qualitative. Bar chart models were used for data that was organized categorically, such as data on TikTok users and readers per genre.

To track the trend of the popularity of genres over time, a stacked surface area model was used. The advantage of this model is that the plot makes the cumulative trend of genres clearer over time, emphasizing the change in individual categories. It also shows a part-to-whole aspect that can be used to compare categories against each other, and the overall trend. However, there are some disadvantages as well. Stacked surface area plots can make it difficult to compare individual categories, as their baselines can get lost. The precise values also get showcased less, and too many categories can clutter the graph. Although these disadvantages are important to take note of, this model was still best to show how genres compared to each other over time.

IV. RESULTS

A. Books Sold Over Time

The first statistic this study looks at is the trend of books sold over time. Three scatter plots were made to show the number of books sold over time; one each for print and e-books, and then one for both (over a range of years included in both sets).

The plot below shows print books sold from 2004-2024. In Figure 1, the trend of print books can be seen to increase linearly up until 2009, where they drop drastically until 2012. From then on, the trend increases steadily overall, with a large jump from 2019 to 2020 and from 2020 to 2021 before it falls again and stabilizes.

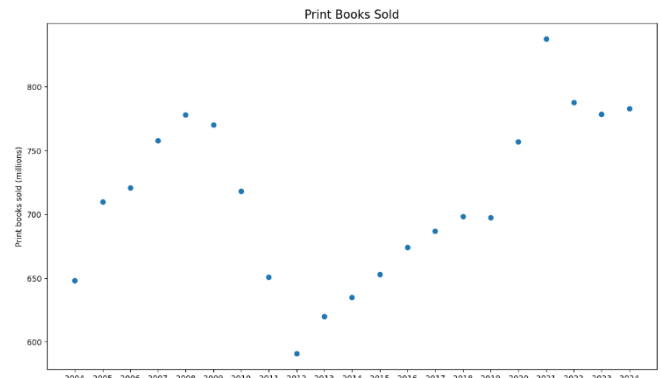


Fig 1. The number of print books sold in the U.S. from 2004-2024

The jump from 2004 to 2005 can most likely be attributed to the release of *Harry Potter and the Half-Blood Prince* by J.K. Rowling, as well as the continuation of the popular adult series *The Da Vinci Code* by Dan Brown.

The two jumps in print books sold from 2019-2021 correlate with the Covid-19 pandemic, where Americans had more time to spend in the house and could not pursue active activities. The details of this trend are explained further below, alongside Figure 3 which displays the total number of books sold in both formats.

Figure 2 displays the trend of e-books sold. An e-book is a book in digital format, such as EPUB, PDF and MOBI. From 2010 to 2013 there is a near linear increase in sales. This includes a large jump in sales from 2010 to 2011. Starting in 2014, the plot shows the beginning of a gradual decrease in e-books sold until 2018, when it starts to go back up.

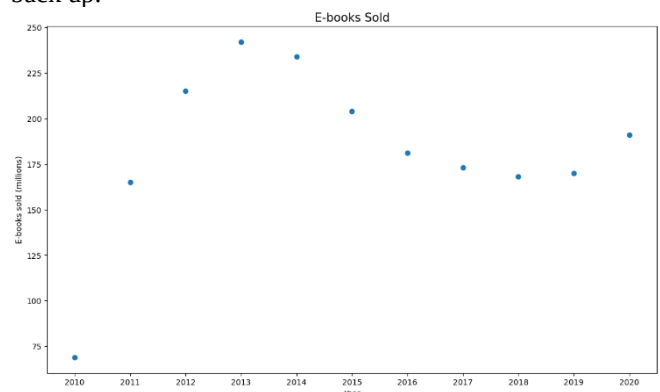


Fig 2. The number of e-books sold in the U.S. from 2010-2020

Comparing Figures 1 and 2, one can see that the drop of print books starting in 2009 happens at the same time as the

rise in e-book sales in 2010. This rise in e-book sales can be largely attributed to the creation of the Amazon Kindle in 2007. In 2010, a feature called Active Content was introduced to Kindles, where additional apps could be installed. However, this feature was removed in 2014 [1]. In Figure 2, the data shows that e-book sales start to decrease, coinciding with the removal of this big feature of the Kindle.

The third scatter plot combines print and e-books to display the overall trend of books sold in the U.S from 2010-2020. While varied, the trend shows an overall increase over time.

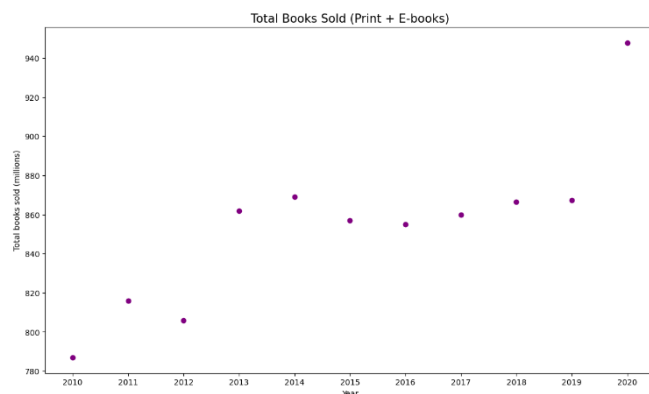


Fig 3. The total number of books sold in the U.S. from 2010-2020

A notable change in trend is in Figure 1, where there is a significant jump in print book sales in 2020. This is also shown in Figure 3, where the number of total book sales jumps over 80 million in one year. A significant event happening in America at the time was the Covid-19 pandemic. While it was harder to go out shopping due to distancing and quarantine, people could still order books shipped to them- but this alone does not explain this rise in sales. However, combining additional data regarding technological changes may shed some light, as a social media app called TikTok begins to gain its massive popularity around this time as well.

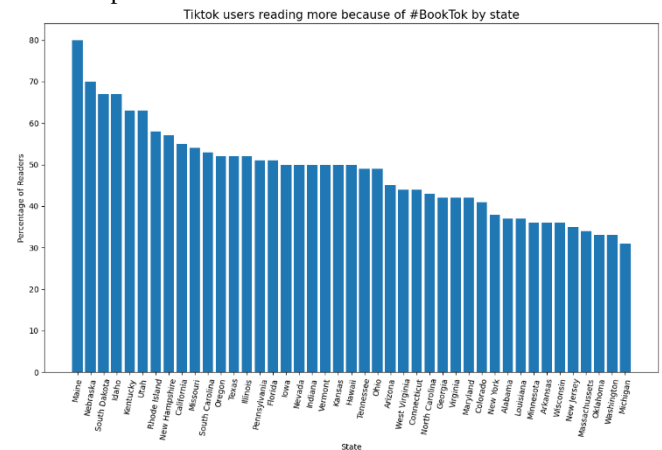
B. TikTok Users Reading More Because #BookTok

TikTok, initially released in 2016, is a short-form video platform and social media app that allows users to post videos. In April of 2020, it surpassed 2 billion downloads and has allowed a multitude of viral trends to take off.

One subcommunity of TikTok, defined by the hashtag #BookTok, is one that revolves around the discussion of books and literature. BookTok creators are primarily teenagers and young women, with a focus on romance, young adult fiction and fantasy genres. This community grew in popularity during the pandemic to the point where publishers reached out to creators for collaborations. BookTok has been linked with increased sales of books (see Figures 2 and 3) and publishing [8]. Some notable books that gained popularity largely due to BookTok are *The Song of Achilles* by Madeline Miller, *The Atlas Six* by Olivie Blake and *Lightlark* by Alex Aster.

In Figure 4, the graph displays data on TikTok users the attributed reading more because of BookTok, per state in May 2023 [7]. A bar chart was used to show the percentage of TikTok users per state reading more because of BookTok. In each state represented, at least 20% of TikTok users stated they were reading more because of #BookTok, and the average percentage was 48.18%, a significant value.

This clearly demonstrates the impact of this community on readership in the United States.



Thirdly, the mid 2000s marks a momentary decrease in publishing, as digital formats start to become popular. This meant publishers and advertisers had to adjust to this changing landscape- which they did, as publishing skyrockets quickly after. This increase in publishing is also a reflection of popular books over time, such as *The Hunger Games* by Suzanne Collins, one of the most popular young adult dystopian series of the century. The trend of genres reflects these popular books and leads to a fast increase in large part due to the speed of information in the digital age.

In terms of genre trends, the plot shows how certain ones became less popular or more popular through the years. From 1920-1980, literature, classics and historical fiction are amongst the more popular genres but fall off heading into the 21st century in favor of fiction adjacent genres. A possible explanation for this is that fiction is a form of mental escapism and in today's world where one can see anything happen anywhere with a click, this retreat is comforting and fun. As such, these genres become more popular.

The paranormal genre makes a notable appearance, a reflection of the publishing of popular books such as *Twilight* by Stephanie Meyer and *City of Bones* by Cassandra Clare. Young adult also increases dramatically at the turn of the century, as young adults become more engaged in online discourse regarding media, and more books come out marketed towards them that gain immense popularity. Romance and contemporary rise in publishing as well, reflecting the female driven consumer market. Together, these trends show how readership has changed over time in both sheer numbers of books read, and how gender has impacted the types of books being published.

B. Genre Distribution by Gender

This last model displays the percentage of readers per genre categorized by male and female readers. This data was taken from a survey in July 2015, and previous findings regarding the connection between BookTok and gender, as well as genre trends show that this data is still relevant today [12]. A bar chart was used in order to display each genre and then the response variable categorized by gender.

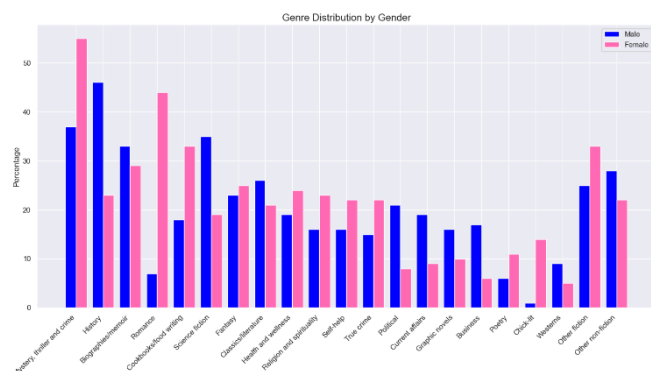


Fig 7. The percentage of respondents categorized by gender per genre

In Figure 7, the data compares male and female readership in 21 genres. Male readers have higher readership in history, science fiction, classics, political current affairs, graphic novels, business, westerns, and other non-fiction genres, making 9. The other 12 genres are led in percentage by female readers, notably including mystery/thrillers/crime, romance, chick-lit and other fiction.

The trend of large female readership in romance and fiction continues here but this data provides additional

insights in gender-genre differences. For instance, there is an 18% difference between female and male readers in the mystery/thrillers/crime genre. The phenomenon of women being more interested in true crime has been documented and researched, and stems from the need to both learn more about what could happen to them in real life and feel safe exploring these topics in a fictional setting [8]. Male readers reading more in history, politics and business reflects how men dominate these fields. Interestingly, this data shows that women prefer fiction whereas men prefer non-fiction for the most part. Research shows that this is based in the gendered socialization of men, where fiction can be seen as “feminine”, as well as the fact that men have been able to use knowledge in books to benefit them in the workplace, which has been accessible to them longer.

V. DISCUSSION

A. Limits of Methodology

This study provided many key insights into the relationship between gender and readership in today's America. However, there are some methodological limitations exposed by this study, such as the difficulty in displaying the data in Figure 5, as the specific trends of genres are lost, and displaying all of them was not visually feasible in a clear way. In addition, the data is not conducive to predictive modeling- as discussed, these trends of both readership and genres are very influenced by current events and societal norms. Predictions made on this data would be under the assumption that American society remains the same, which is incorrect.

B. Limitations on Key Insights

Literature is a large spanning topic, specifically timewise. Even looking at only the United States, there is 250 years' worth of books. However, the data does not go that far back in terms of readership trends. This study is also limited by datasets such as Genre Popularity by Gender and TikTok users reading more because of BookTok, as these are from one-time surveys and do not track changes in those trends over time. Additionally, data regarding e-books [3] is not comprehensive- there are many places online where one can download e-books for free, but this dataset only tracks e-books sold.

While an increase in book sales may imply a rise in readership, this is not necessarily the case. Since the popularity of online book communities, the trend of book collecting has risen, meaning that people who buy books may only do so due to the consumeristic nature of this trend. In fact, research done by the National Endowment for the Arts shows that overall readership has been declining even though there is a gender gap between male and female readers [9]. In 2012, 54.6% of U.S. adults said they had read a book. In 2017, 52.7% of U.S. adults reported reading a book and in 2022, 48.5% said they had read a book. It is important to note however, that these statistics do not include audiobooks, which are listenable versions of books. For example, considering audiobooks in 2022, 51.9% of adults had read a book (18.5% having listened) [9]. This still shows a decrease in reading, but a more gradual one. However, sales still reflect the gender gap in reading- avid readers (those who read on their own time) are still female in majority. A study done by author Anne Bogel showed that as of 2015, 88% of avid readers were female, and 80% of book purchases were made by women [6]. While this

does not mean that the rate of actual reading perfectly corresponds to book sales, there is still a strong correlation between women reading more than men.

C. Suggestions on Future Research

Future research could delve much deeper into historical events that impact reading and publishing that can explain gendered reading trends in more detail. There are many more events throughout U.S. history that impact readership both genre and gender wise than mentioned in this study. Additionally, more datasets spanning over more years could be incorporated to analyze more of the United States over time. Further work specifically on the trend of genres over time to visualize more of that data could be done to provide more insights.

VI. CONCLUSION

This research reveals the many sociocultural events and factors that shape the relationship between reading and gender. One major factor is technological advancement, where reading online has kept people reading due to its accessibility, even with a gradual decrease in reading overall [9]. While many people of different backgrounds and demographics can read online, studies show that women are in the majority when it comes to owning Kindles and e-readers [10]. This correlates to the rise of majority female reading genres such as romance and fiction genres. Online book communities such as BookTok also play a large role in the literary world today, and as previously discussed, women are the majority in these types of communities which naturally shapes the trend of genres in the 21st century.

Historical and sociocultural events also have a large impact on readership and specifically gendered reading. The changing landscape of the 20th century in particular impacted book publishing and readership heavily. New technologies, world wars and a multitude of civil rights movements in the United States impacted both what Americans were writing and reading, as well as who was reading. The popularity of genres can be cyclical; take *The Hunger Games*, which started a trend of dystopian young adult series such as *Divergent* by Veronica Roth. This cycle reflects what people find interesting, as well as relevant. Many forms of work of all medias gain popularity in large part because consumers find their thematic content parallel to the real world. A famous example of this is *Star Wars*, which was inspired by WW2, the Vietnam War and the presidency of President Nixon, and continues to depict these real-world parallels in a sci-fi environment.

In the 21st century, the reading landscape is vastly different from previous years. As women became more able to pursue an education and work in 20th century America, female literacy rates went up and therefore both overall readership and female read genres did as well [2]. Women now read more than men on average and the way books are perceived is heavily influenced by online posting and interaction, where mostly female participants and consumers drive the market in a much stronger way than ever before. In specific regards to gender, there is discussion to be had about what this gender gap means. In the United States, girls and women outperform their male counterparts in academic achievement and find more enjoyment out of reading. There are many societal factors that play into this, such as gendered societal norms that see reading as feminine and how online reading spaces are often female dominated. Literature of all kinds is important for everyone- even fiction helps us study the real world

through the parallels made by a fake one. To help close this gender gap, reading curriculums in schools can be adjusted and the destigmatization of stereotypically female genres is necessary. This will also help more people read more books as opposed to participating in consumeristic book buying or only consuming other media.

In addition to the relationship between gender and readership, there are other demographic factors that correlate with reading. Economic status and racial identities also have a relationship with readership as well as gender. A further exploration of these connections would yield more light on the state of reading in today's America.

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